

Urchade Zaratiana

Member of Technical Staff at Fastino Labs · PhD in Computer Science

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I build small language models that return structured outputs quickly and reliably. Creator of the [GLiNER](#) family: encoder models for zero-shot extraction, classification and safety, run in production at NVIDIA, Meta, Airbnb and across the open-source ecosystem, and matching decoder baselines 23 to 90 times their size. PhD on structured prediction; ATALA Best Thesis Award 2025.

EXPERIENCE

Member of Technical Staff *Fastino Labs*

Palo Alto, CA (remote) · Jul 2024 to Present

- First hire; lead the Data Structuring team on structured prediction: fast inference, structured prediction and small language models.
- Own small-language-model architecture development and every open-source release: the GLiNER family, spanning [GLiNER2](#) (multi-task schema-driven extraction), [GLiGuard](#) (LLM safety) and [GLiNER2-PII](#) (multilingual PII), over 35M downloads to date on Hugging Face and around 1M per month.
- Architect of the Fastino Labs AutoML platform: synthetic data generation, training pipelines, and the continual-improvement loop published as [Pioneer Agent](#) (senior author).
- Shipped task-specific customer models on the same stack: e-commerce query parsing, a custom RAG architecture, and fast non-autoregressive JSON extraction.

Research Scientist *FI Group*

Paris, France · Dec 2020 to Jul 2024

- Industrial research scientist under a CIFRE contract, held jointly with the PhD at LIPN.
- Built the information-extraction models behind [Neophi](#), FI Group's platform for exploring innovation and funding data: entity and keyphrase extraction on scientific text, knowledge-graph construction, taxonomy induction, and cross-lingual search and QA.

Research Intern *Astek*

Paris, France · Feb to Jul 2020

- Adapted pretrained encoders (BERT) to web article recommendation. Advisor: Alexandre Pascault.

SELECTED PUBLICATIONS

1. [Pioneer Agent: Continual Improvement of Small Language Models in Production](#). Fastino Labs; **Zaratiana** (senior author). *arXiv* 2026.
2. [GLiNER2-PII: A Multilingual Model for Personally Identifiable Information Extraction](#). **Zaratiana**, Lewis, Hurn-Maloney. *arXiv* 2026.
3. [GLiGuard: Schema-Conditioned Classification for LLM Safeguard](#). **Zaratiana**, Newhauser, Hurn-Maloney, Lewis. *COLM* 2026.

4. GLiNER2: An Efficient Multi-Task Information Extraction System with Schema-Driven Interface. **Zarathiana**, Pasternak, Boyd, Hurn-Maloney, Lewis. *EMNLP 2025, System Demonstrations*.
5. GLiNER: Generalist Model for Named Entity Recognition using Bidirectional Transformer. **Zarathiana**, Tomeh, Holat, Charnois. *NAACL 2024*.

Full list: [Google Scholar](#).

EDUCATION

PhD in Computer Science *Université Sorbonne Paris Nord*

Paris, France · 2021 to 2024

- Thesis at the Laboratoire Informatique de Paris Nord (LIPN), RCLN team: entity and relation extraction with graph-structured prediction. Advisors: Dr. Nadi Tomeh and Prof. Thierry Charnois.
- Structured decoding and structured losses, including [autoregressive text-to-graph extraction](#) (AAAI 2024), [Filtered Semi-Markov CRF](#) (EMNLP 2023) and [GNNer](#) (ACL 2022); the line of work GLiNER grew out of.

MSc Computer Science *Johannes Kepler University*

Linz, Austria · 2019 to 2020

- Deep learning and neural networks, NLP with deep learning, probabilistic models, information retrieval, deep learning in life sciences.
- Project: abstractive text summarization for French. Advisor: Dr. Navid Rekab-saz.

Software Engineering Degree *ISEP*

Paris, France · 2017 to 2020

- Algorithmics, artificial intelligence, web programming. Project: home-automation website. Advisor: Dr. Rafael Angarita.

AWARDS

ATALA Best Thesis Award

2025

Awarded annually by ATALA for the most outstanding doctoral research in natural language processing in France.

SKILLS

Research

Structured prediction · information extraction · constrained and non-autoregressive decoding · zero-shot and schema-driven modelling · LLM safety and PII

Modelling

Encoder architectures for small language models · pretraining and finetuning · synthetic data generation · efficient CPU and GPU inference · evaluation and continual improvement in production

Engineering

Python · PyTorch · Transformers · distributed training · AutoML pipelines · open-source library design and maintenance

Leading

Founding-engineer scope · building and directing a research team · turning research into shipped customer models

Languages

French (native) · English (fluent) · Malagasy